

Smart Fall Detection System

CS61066Term Project

ARCHITECTURE AND PROTOCOLS FOR INTERNET OF THINGS

CS61066

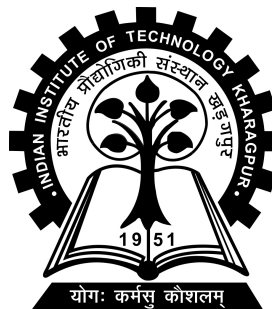
by

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Under the supervision of

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Department of Computer Science and Engineering

Indian Institute of Technology Kharagpur

Autumn Semester, 2025-26

11th November 2025

DECLARATION

I certify that

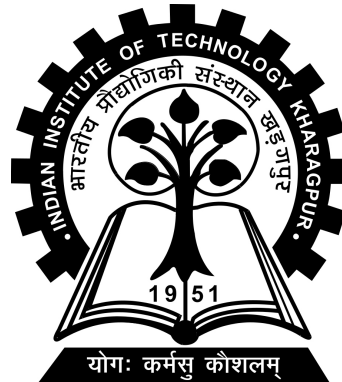
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Date: 11th November 2025

Place: Kharagpur

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CERTIFICATE

This is to certify that the project report entitled “Smart Fall Detection System” submitted by Adhithya and Tejaraam (Roll No. 22CS30019 and 22CS30059) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2025-26.

Date: 11th November 2025

Place: Kharagpur

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Abstract

Name of the student: **Adhithya and Tejaraam** Roll No: **22CS30019 and 22CS30059**

Degree for which submitted: **Bachelor of Technology**

Department: **Department of Computer Science and Engineering**

Thesis title: **Smart Fall Detection System**

Thesis supervisor: **Professor Sudip Misra**

Month and year of thesis submission: **11th November 2025**

A Fall-Detection Device basically monitors the accelerometer and gyroscope readings of the person to which it was attached and tries to identify the fall pattern from the statistical data, which it derives from the sensor data. For analysing those, we will use a trained ML model deployed in the cloud. This device is not only capable of finding the fall patterns, but it can also collect the movement data in terms of statistical features of accelerometer and gyroscopic readings for future use, which can be further used to train the deployed model after labelling.

The Device was intended to be multi-tasking. The device has two different modes, namely Data-Collection Mode, Data-Prediction Mode, for serving two unique functionalities. Data-Collection Mode: Our Device will collect the data from the sensors and compute the statistical features like mean, max, min and variance for each of the 6 sensor parameters [ax,ay,az,gx,gy,gz] and send to the backend server to store for future labelling purposes. Data-Prediction Mode: Uses the trained ML model deployed in the cloud for real-time fall prediction.

Keywords - Wearable Sensor, Accelerometer, Gyroscope, Machine Learning, Random Forest, IoT, Fall Detection

Acknowledgement

I would like to express my sincere gratitude to my supervisor, Prof. Sudip Misra, for his constant guidance, valuable suggestions, and encouragement throughout this project. I am also thankful to our teaching assistant, Mr. Biswajit Ghosh, for his support and helpful feedback during the development of this work.

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Abbreviations

ML	M achine L earning
IoT	I nternet o f T hings
RF	R andom F orest
XGBoost	E xtrême G radient B oosting

Chapter 1

Introduction

1.1 Introduction

A Fall-Detection Device basically monitors the accelerometer and gyroscope readings of the person to which it was attached and tries to identify the fall pattern from the statistical data, which it derives from the sensor data. For analysing those, we will use a trained ML model deployed in the cloud . This device is not only capable of finding the fall patterns, but it can also collect the movement data in terms of statistical features of accelerometer and gyroscopic readings for futur usee, which can be further used to train the deployed model after labelling

1.2 Motivation

Many Elderly people will die due to unexpected fall accidents, to make sure we have continuous monitoring of the movement, what if we have a device that analyses the movement patterns, continuously identifies the fall accidents immediately within a fraction of second, alerting the caretakers.

1.3 Contribution of Project

This project was made into three different sections Frontend, Backend and Hardware.

The Frontend and hardware sections are looked after by the V.Tejaraam which includes creation of visually appealing dashboard and interactive labeling page along with hardware simulation and connection.

The system architecture and backend server design along with model creation was looked after by D.Adhithya, he design the endpoints, end to end communication and model for identifying the fall pattern, all the connectivity issues, debugging, logs and managing database was covered in the backend server design itself.

Chapter 2

Features

The Device was intended to be multi-tasking. The device has two different modes, namely “Data-Collection Mode”, “Data-Prediction Mode”, for serving two unique functionalities.

2.1 Data-Collection Mode

Our Device will collect the data from the sensors and compute the statistical features like mean, max, min and variance for each of the 6 sensor parameters [ax,ay,az,gx,gy,gz] and send to the backend server to store for future labelling purposes.

2.2 Data-Prediction Mode

Device will collect and compute the statistical features of sensor data and send the server and server will use the saved ML model to do the prediction and will post a alert message in a MQTT Topic if there is a fall , our device wil give a alert if any message posted in that MQTT topic.

2.3 Real-time Frontend Dashboard and Retraining Feature

This System also provides us a real-time frontend dashboards and also retraining feature of the ML model in the form of button.

2.3.1 Analytics Page

It Provides a real-time dashboard which visualises the last 10 min status of the person in which we can observe the whether there is a fall action noted or not.

2.3.2 Training Page

All the unlabelled data points from the database are fetched into single table and a labelling option will be made available once a datapoint is labelled it will be automatically updated in the database and in the frontend as well without need of refreshing the page.

2.3.3 Model Retraining

This Button Enables us to re-train the model, on the new dataset which contains all the labeled data.

Chapter 3

System Methodology

3.1 System Components and Architecture

Smart-Fall Detection System is built on a multi-layer IoT architecture that connects everything efficiently. It's designed such that data flows easily from the IoT device to the cloud application, and then sends back an alert right away. Using communication methods that save battery power, this creates a reliable and practical system that watches over the patient all the time.

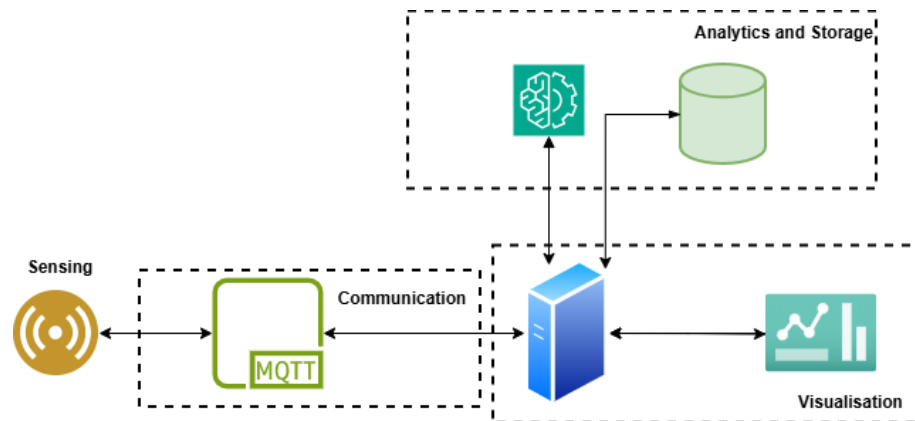


FIGURE 3.1: Overall IoT Architecture of Smart-Fall Detection System

TABLE 3.1: IoT System Layers and Components

Layer	Components	Description
Sensing	MPU6050	Collects accelerometer and gyroscope raw data.
Edge Processing	ESP32 microcontroller	Collects 100 samples from the MPU6050 sensor and computes statistical features (mean, max, min, variance) for all six variables (Ax , Ay , Az , Gx , Gy , Gz).
Communication	Wi-Fi, MQTT protocol	MQTT and Wi-Fi are used for power-efficient publish-subscribe communication between the IoT node and the backend server.
Analytics	Random Forest Model	A pretrained Random Forest model with 24 features is deployed on the backend server to identify fall patterns.
Actuation	LED, Buzzer	Actuates the fall alert signal.
Application	React Frontend Dashboard	Provides real-time monitoring, prediction analytics, and a dashboard for data labeling and model retraining.

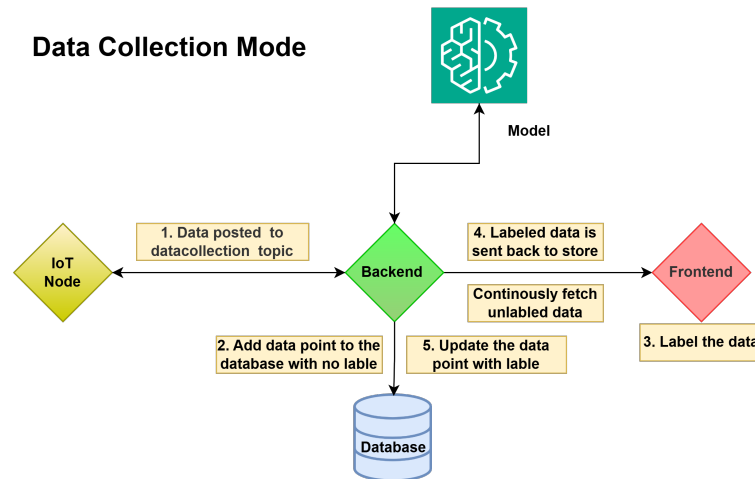


FIGURE 3.2: Data Collection Mode

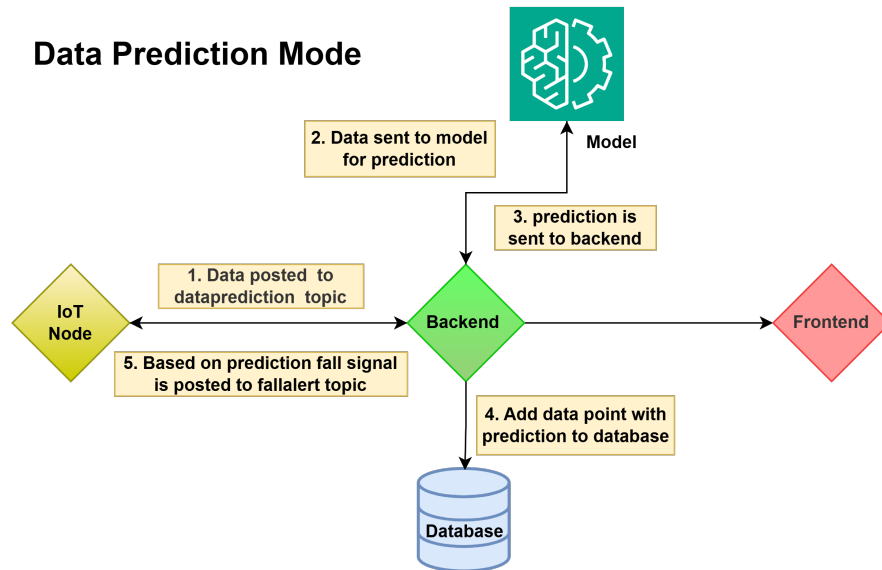


FIGURE 3.3: Data Prediction Mode

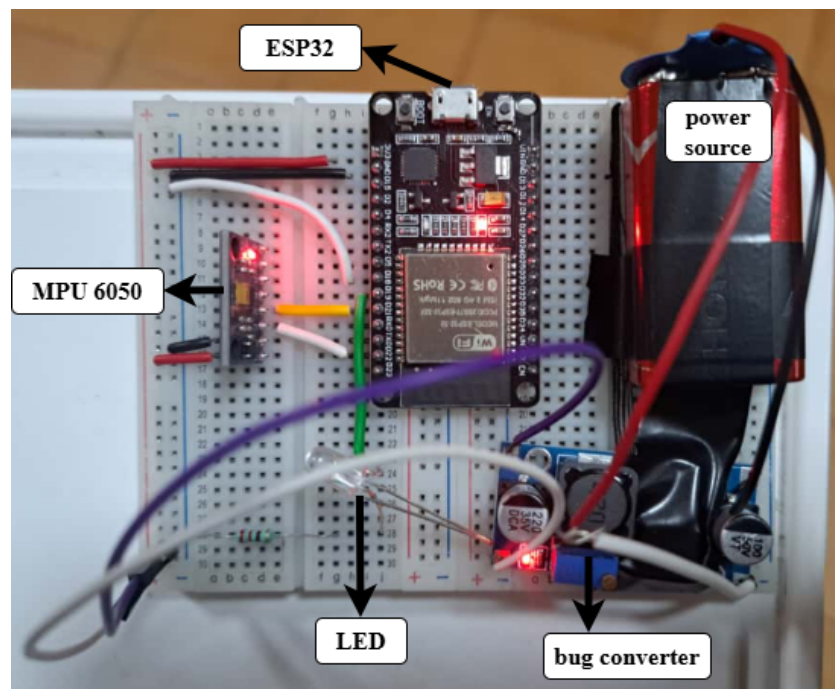


FIGURE 3.4: Hardware Setup

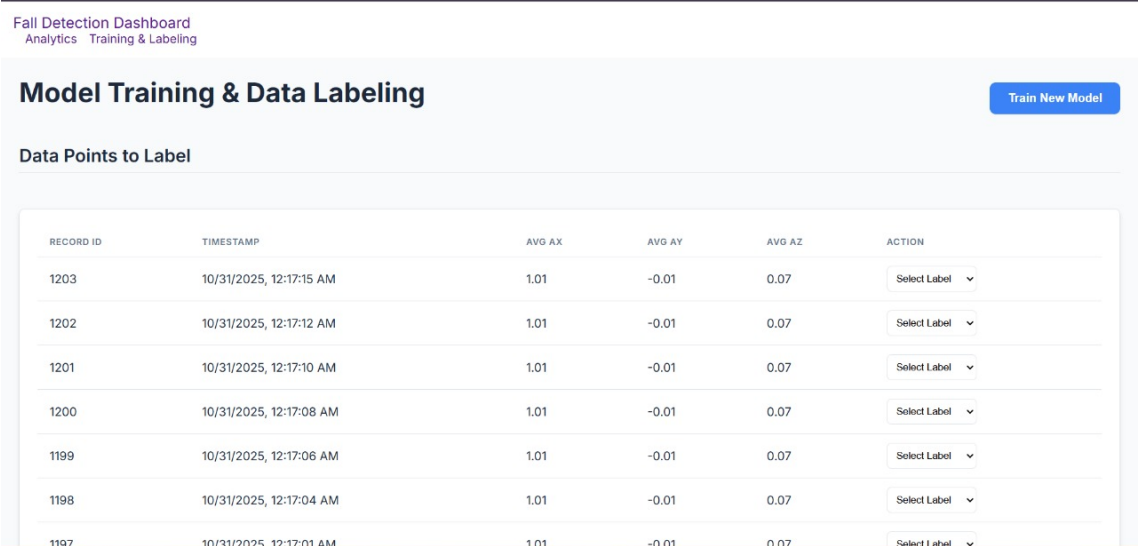


FIGURE 3.5: React Frontend Dashboard — Data Labeling

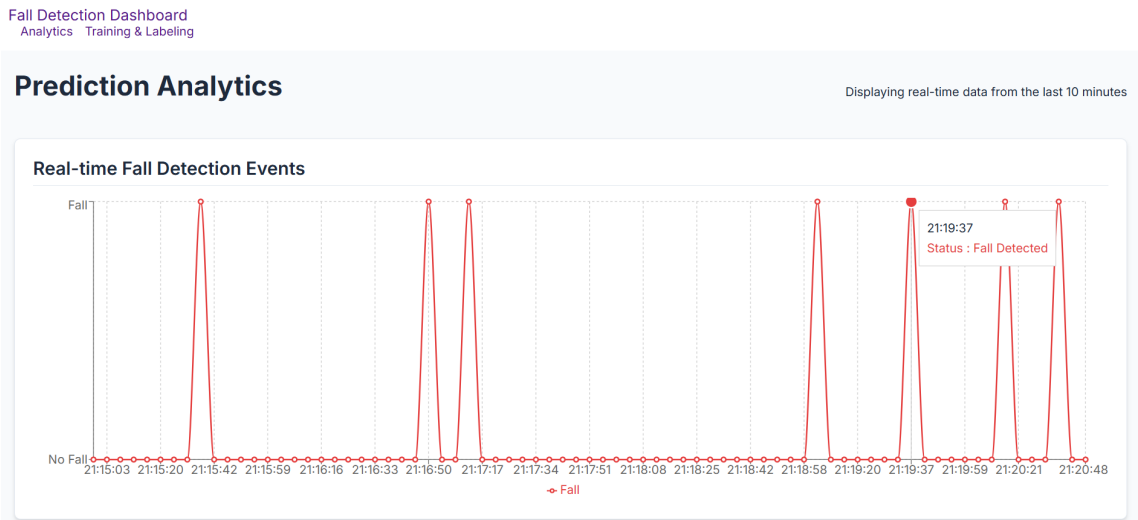


FIGURE 3.6: React Frontend Dashboard — Analytics View

Chapter 4

Experimental Setup

4.1 Dataset Preparation

4.1.1 Custom Data Collection

Initially we used sisfall dataset but the sensor used to create this dataset is not same as our sensor. So we manually collect the data and labelled the data with our sensor.

We need to tag fall or nofall manually for every datapoint and initial most of the times when experiment is conducted if we connected module with battery , battery used to get disconnected so we connected usb to the ESP32 and collected the data.

4.1.2 Selection of ML Model

Since our dataset was prepared manually due to constrain of resources we are only able to collect 1000 datapoints, so due to lack of knowledge about different models performance on small dataset we were not able to prioritise which model to use initially. So we trained regression model, decision tree, random forest, Xg boost, neural network models and decided based on the accuracy of model.

4.1.3 Hardware Setup

The hardware device we have designed typically contains four different sections.

Power Source : We have used a 9V battery and a LM2596 Buck convertor to step down to the 3.3V power supply.

MPU6050 Sensor: This is the Sensor which collects the accelerometer and Gyroscopic data of the person ,it send the data from the SDA and SCL to the D21 and D22 Pins of the ESP32.

ESP32 Microcontroller: This is the Central Processing Unit of the hardware device ,which collects the sensor data processes it and determine where to post based on the input signal from the MODE Pin which is D18,and also on receiving the alertsignal on the ‘fallalert’ topics it will make the D19 pin high to actuate LED.

Actuator (LED) : we have used an LED light for actuation ,Whenever there is a fall detected it will be actuated by ESP32.

Chapter 5

Challenges

5.1 Challenges we faced

1. Finding proper dataset for our usecase: Initially we used sisfall dataset but the sensor used to create this dataset is not same as our sensor. So we manually collect the data and labelled the data with our sensor.
2. Selection of ml model: Since our dataset was prepared manually due to constrain of resources we are only able to collect 1000 datapoints, so due to lack of knowledge about different models performance on small dataset we were not able to prioritise which model to use initially. So we trained regression model, decision tree, random forest, Xg boost, neural network models and decided based on the accuracy of model.
3. Preparing our own dataset: We need to tag fall or nofall manually for every datapoint and initial most of the times when experiment is conducted if we connected module with battery , battery used to get disconnected so we connected usb to the esp32 and collected the data.

-
4. Wifi connectivity and Power issue: While the system worked well when powered via a USB adapter, achieving stable operation when disconnected and running entirely on a battery unit proved difficult
 5. Frequent refreshing of page issue for frontend dashboard

Chapter 6

Conclusion and Further Work

6.1 Conclusion

In conclusion, the Fall-Detection Device helps keep track of a person's movements using data from the accelerometer and gyroscope. It can quickly find out if a fall has happened by using a trained machine learning model stored in the cloud. Along with detecting falls, the device also saves movement data in the form of useful numbers. This data can later be labelled and used to retrain the model, making the system smarter and more accurate over time.

6.2 Future Work

Future work involves addressing the challenges identified in the current system, specifically:

1. Enhancing stability and robustness of the system while running entirely on a battery unit for long time, like using batteries which can charge.
2. Collecting a larger, more diverse dataset to train more robust ML models.

-
3. As this is a prototype we actuated using led, insted we can send this actuation signal as notification to caretaker's mobile.

Bibliography